Week 2 - Linear Regression

Mock Questions + Full Answers

Practice the notation, formulas, methods, and interpretation from the lecture notes.

How to use this sheet. Try all questions before looking at the answers. The questions are ordered from basic notation to full fitting methods. For calculation questions, write the formula first, substitute values second, and only then simplify.

Questions

A. Core notation and ideas

Q1. In a dataset $(x_1, y_1), \ldots, (x_N, y_N)$, explain what each of the following means: N, i, x_i, y_i, f_{true}, and ε_i in

$$y_i = f_{\text{true}}(x_i) + \varepsilon_i.$$

Which quantities are observed and which are unknown?

Q2. For the affine predictor

$$f_\theta(x) = b + ax, \qquad \theta = \begin{pmatrix} b \\ a \end{pmatrix},$$

state what a, b, and θ represent. Which quantities are fixed during fitting, and which quantities vary?

Q3. A model predicts $\hat{y} = 7.5$ while the target is $y = 9$.

(a) Compute the residual $r = \hat{y} - y$.

(b) Compute the absolute loss $|r|$.

(c) Compute the squared loss r^2.

(d) Is this an overprediction or underprediction?

Q4. Explain the difference between the individual squared loss

$$\ell_{\text{sq}}(\hat{y}, y) = (\hat{y} - y)^2$$

and the empirical squared loss

$$\mathcal{L}_{\text{sq}}(\theta) = \sum_{i=1}^{N} (f_\theta(x_i) - y_i)^2.$$

Q5. Why does squared loss penalise large errors more strongly than absolute loss? What practical advantage of squared loss is emphasised in the notes?

B. Using the simple fitting formulas

For Questions 6–10 use the dataset

$$\boxed{x = (-2, -1, 0, 1, 2),\ y = (-2, 0, 1, 3, 6)}$$

so $N = 5$.

Q6. Find the best **constant predictor** $f_b(x) = b$ under squared loss. Compute b^* and explain what familiar statistic it equals.

Q7. For the fitted constant from Q6, compute all five residuals and the total squared loss.

Q8. Fit a **line through the origin**, $f_a(x) = ax$. Use

$$a^* = \frac{\sum_{i=1}^N x_i y_i}{\sum_{i=1}^N x_i^2}.$$

Compute a^*, then write the fitted predictor.

Q9. For the slope-only predictor from Q8, compute the five predictions, the five residuals, and the total squared loss.

Q10. Fit the full affine predictor $f(x) = b + ax$. Use

$$a^* = \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^N (x_i - \bar{x})^2}, \qquad b^* = \bar{y} - a^* \bar{x}.$$

Find $\bar{x}$, $\bar{y}$, a^*, and b^*. Which of the three families from Q6, Q8 and Q10 gives the smallest training loss on this dataset?

C. Matrix least squares

Q11. Consider the three observations

$$(0, 1), \quad (1, 2), \quad (2, 5).$$

Construct the design matrix X, parameter vector θ, and target column $\mathbf{y}$ for the affine model $f(x) = b + ax$.

Q12. Using the data from Q11, compute

$$X^T X \qquad \text{and} \qquad X^T \mathbf{y}.$$

Then write the normal equations

$$X^T X \, \theta = X^T \mathbf{y}.$$

Q13. Solve the normal equations from Q12 to obtain b^* and a^*. Then compute the fitted predictions and the empirical squared loss.

Q14. State the matrix formula for the least-squares solution when $X^T X$ is invertible. Give the dimensions of X, θ, $\mathbf{y}$, and $X^T X$ when there are N observations and d input features plus an intercept.

Q15. What happens if $X^T X$ is not invertible? Does that mean the normal equations have no solution? Answer according to the notes.

D. Nonlinear features

Q16. Define the quadratic feature map

$$\phi(x) = (1, x, x^2)$$

and parameter vector

$$\theta = \begin{pmatrix} 1 \\ 2 \\ 0.5 \end{pmatrix}.$$

Write the resulting predictor $f_\theta(x) = \phi(x)\theta$, then compute its predictions at $x = -1, 0, 2$.

Q17. For inputs $x_1 = -1$, $x_2 = 0$, $x_3 = 2$, construct the transformed design matrix Φ for the features $(1, x, x^2)$. State the matrix form of the squared loss and the normal equations using Φ.

Q18. Why is $f(x) = b + w_1 x + w_2 x^2$ still called a **linear regression model**, even though it is quadratic in x? Contrast this with $f_w(x) = \sin(wx)$.

E. Gradient descent

Q19. State the general gradient-descent update and explain the meaning of t and η:

$$\theta^{(t+1)} = \theta^{(t)} - \eta \nabla \mathcal{L}(\theta^{(t)}).$$

Why do we subtract the gradient?

Q20. For the affine squared-loss model, write the matrix gradient

$$\nabla \mathcal{L}_{\text{sq}}(\theta)$$

and the corresponding gradient-descent update in terms of X, θ, and $\mathbf{y}$.

Q21. Use the lecture's five-observation example

$$x = (-2, -1, 0, 1, 2), \qquad y = (-1.5, 0.5, 1, 2.5, 7.5),$$

starting from

$$\theta^{(0)} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \qquad \eta = 0.04.$$

The initial gradient is

$$\nabla \mathcal{L}(\theta^{(0)}) = \begin{pmatrix} -20 \\ -40 \end{pmatrix}.$$

Compute $\theta^{(1)}$ and the new predictor.

Q22. A student increases the learning rate substantially and sees the loss increase from one iteration to the next. Is this necessarily a coding error? Explain.

Q23. Explain the difference between **fitting loss on the observed data** and **generalisation to unseen data**. Does a very small training loss guarantee good unseen predictions?

Answers

A. Core notation and ideas

A1. N is the number of observations. i is the observation index. x_i is the input for observation i, and y_i is its observed target. f_{true} is the unknown underlying relationship and ε_i is the unknown noise for observation i. We observe (x_i, y_i); we do not observe f_{true} or the individual ε_i.

A2. a is the slope, b is the intercept (bias), and θ collects the adjustable parameters. During fitting, the observed inputs and targets stay fixed; the parameters vary.

A3.

$$r = 7.5 - 9 = -1.5, \qquad |r| = 1.5, \qquad r^2 = 2.25.$$

Because $\hat{y} < y$, this is an underprediction.

A4. ℓ_{sq} measures the penalty for one prediction-target pair. $\mathcal{L}_{\text{sq}}(\theta)$ uses the parameters to make all predictions and sums the individual squared losses over the dataset. It is the objective we minimise during fitting.

A5. Squaring makes the penalty grow quadratically: doubling an error multiplies its squared penalty by four. Squared loss is also differentiable everywhere, which makes derivative-based optimisation straightforward.

B. Using the simple fitting formulas

A6. The best constant under squared loss is the mean of the targets:

$$b^* = \bar{y} = \frac{-2 + 0 + 1 + 3 + 6}{5} = \frac{8}{5} = 1.6.$$

A7. With $b^* = 1.6$, the residuals are

$$3.6,\ 1.6,\ 0.6,\ -1.4,\ -4.4.$$

Hence

$$\mathcal{L} = 3.6^2 + 1.6^2 + 0.6^2 + (-1.4)^2 + (-4.4)^2 = 37.2.$$

A8.

$$\sum x_i^2 = 4 + 1 + 0 + 1 + 4 = 10,$$

$$\sum x_i y_i = (-2)(-2) + (-1)(0) + 0(1) + 1(3) + 2(6) = 19.$$

Therefore

$$a^* = \frac{19}{10} = 1.9, \qquad f(x) = 1.9x.$$

A9. Predictions:

$$-3.8,\ -1.9,\ 0,\ 1.9,\ 3.8.$$

Residuals $(\hat{y} - y)$:

$$-1.8,\ -1.9,\ -1,\ -1.1,\ -2.2.$$

Loss:

$$3.24 + 3.61 + 1 + 1.21 + 4.84 = 13.9.$$

A10. Since $x = (-2, -1, 0, 1, 2)$,

$$\bar{x} = 0, \qquad \bar{y} = 1.6.$$

With $\bar{x} = 0$,

$$a^* = \frac{\sum x_i(y_i - \bar{y})}{\sum x_i^2} = \frac{19}{10} = 1.9, \qquad b^* = \bar{y} - a^*\bar{x} = 1.6.$$

Thus

$$f(x) = 1.6 + 1.9x.$$

Predictions are $-2.2, -0.3, 1.6, 3.5, 5.4$, residuals are $-0.2, -0.3, 0.6, 0.5, -0.6$, and the loss is

$$0.04 + 0.09 + 0.36 + 0.25 + 0.36 = 1.10.$$

So the full affine family gives the smallest loss: $1.10 < 13.9 < 37.2$.

C. Matrix least squares

A11.

$$X = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix}, \qquad \theta = \begin{pmatrix} b \\ a \end{pmatrix}, \qquad \mathbf{y} = \begin{pmatrix} 1 \\ 2 \\ 5 \end{pmatrix}.$$

The first column of X handles the intercept.

A12.

$$X^T X = \begin{pmatrix} 3 & 3 \\ 3 & 5 \end{pmatrix}, \qquad X^T \mathbf{y} = \begin{pmatrix} 8 \\ 12 \end{pmatrix}.$$

Therefore

$$\begin{pmatrix} 3 & 3 \\ 3 & 5 \end{pmatrix} \begin{pmatrix} b \\ a \end{pmatrix} = \begin{pmatrix} 8 \\ 12 \end{pmatrix}.$$

A13. The equations are

$$3b + 3a = 8, \qquad 3b + 5a = 12.$$

Subtracting gives $2a = 4$, so $a^* = 2$. Then $3b + 6 = 8$, so

$$b^* = \frac{2}{3}.$$

Predictions:

$$\frac{2}{3}, \frac{8}{3}, \frac{14}{3}.$$

Residuals:

$$-\frac{1}{3}, \frac{2}{3}, -\frac{1}{3}.$$

Loss:

$$\frac{1}{9} + \frac{4}{9} + \frac{1}{9} = \frac{2}{3}.$$

A14. If $X^T X$ is invertible,

$$\boxed{\theta^* = (X^T X)^{-1} X^T \mathbf{y}}.$$

With N observations and d input features plus an intercept:

$$X \in \mathbb{R}^{N \times (d+1)}, \quad \theta \in \mathbb{R}^{(d+1) \times 1}, \quad \mathbf{y} \in \mathbb{R}^{N \times 1},$$

$$X^T X \in \mathbb{R}^{(d+1) \times (d+1)}.$$

A15. The inverse formula is unavailable. According to the notes, the normal equations still have solutions, but the fitted coefficients are not unique. Their solutions still give the same fitted predictions on the observed inputs.

D. Nonlinear features

A16.

$$f_\theta(x) = 1 + 2x + 0.5x^2.$$

Therefore

$$f(-1) = 1 - 2 + 0.5 = -0.5,$$

$$f(0) = 1,$$

$$f(2) = 1 + 4 + 2 = 7.$$

A17.

$$\Phi = \begin{pmatrix} 1 & -1 & 1 \\ 1 & 0 & 0 \\ 1 & 2 & 4 \end{pmatrix}.$$

The squared loss is

$$\mathcal{L}(\theta) = \|\Phi\theta - \mathbf{y}\|_2^2,$$

and the normal equations are

$$\Phi^T\Phi\theta = \Phi^T\mathbf{y}.$$

A18. The model is linear in the adjustable coefficients b, w_1, w_2, even though it is nonlinear in the input x. The features $1, x, x^2$ are fixed before fitting. In contrast, $\sin(wx)$ is nonlinear in the adjustable coefficient w, so it cannot be written as a fixed feature multiplied linearly by w.

E. Gradient descent

A19.

$$\theta^{(t+1)} = \theta^{(t)} - \eta\nabla\mathcal{L}(\theta^{(t)}).$$

Here t is the iteration index and $\eta > 0$ is the learning rate/step size. We subtract the gradient because the gradient points in the direction of fastest local increase, so the negative gradient points locally downhill.

A20.

$$\boxed{\nabla\mathcal{L}_{\text{sq}}(\theta) = 2X^T(X\theta - \mathbf{y})}.$$

Hence

$$\boxed{\theta^{(t+1)} = \theta^{(t)} - 2\eta X^T(X\theta^{(t)} - \mathbf{y})}.$$

Equivalently, compute residuals $\mathbf{r}^{(t)} = X\theta^{(t)} - \mathbf{y}$, then update

$$\theta^{(t+1)} = \theta^{(t)} - 2\eta X^T\mathbf{r}^{(t)}.$$

A21.

$$\theta^{(1)} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} - 0.04\begin{pmatrix} -20 \\ -40 \end{pmatrix} = \begin{pmatrix} 0.8 \\ 1.6 \end{pmatrix}.$$

Therefore the new predictor is

$$\boxed{f(x) = 0.8 + 1.6x}.$$

A22. No. The notes explicitly warn that a step size that is too large can increase the loss. Gradient descent only guarantees a local decrease for a sufficiently small step based on the first-order approximation.

A23. Training/fitting loss measures performance on the observations included in the empirical loss. Generalisation refers to performance on unseen observations. A very small training loss does not by itself guarantee good unseen predictions; underfitting and overfitting are separate issues from the fitting calculation.

Methods you should now be able to use: residual calculation; individual and empirical squared loss; best constant; slope-only fit; affine fit; design matrix construction; normal equations; closed-form least squares; nonlinear feature expansion; gradient calculation; and one gradient-descent update.